



BITS Pilani
DIGITAL

READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 4

Functions and Workings of Secondary Markets



STUDY NOTES · EDITION 1.0 · WEEK 4

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How to use these notes



These notes cover the whole of Week 4: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A covers where trading actually happens — exchanges, order types and the over-the-counter markets that carry more volume than either. Part B examines liquidity and market efficiency, the two properties that determine whether a market works. Part C follows a trade from order to legal finality. Read the parts in order the first time; each assumes the last.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

What each topic contains

Every topic follows the same rhythm, so you always know where to look.

Section	What it gives you
In one line	The whole idea compressed to a single sentence
Why this matters	Where the concept sits in the course, and why it earns your attention
The core idea	The full explanation, developed step by step
Figure	A diagram or comparison that makes the idea visible
Callouts	Key ideas, worked examples, pitfalls and exam pointers
Quick recap	Five bullets to revise from

Callouts used in this document

KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

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A Where trading happens

1	Stock exchanges: how orders become one price
2	The four order types
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B Liquidity and efficiency

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6	Price discovery and market efficiency

C Trading and settlement

7	Market structure and trading costs
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9	Clearing and settlement

CONSOLIDATION

Module summary, by learning objective

Glossary and self-check

Module overview



This module opens the machinery of the secondary market. It begins with the venues — organised exchanges, the four order types investors use to speak to them, and the far larger over-the-counter markets that carry government debt and foreign exchange. It then turns to the two properties that decide whether a market functions: liquidity, and the efficiency with which prices absorb information. It closes by following a single trade from execution through to irrevocable settlement.

One thread runs through all nine topics: **the structure of a market determines its outcomes**. Where a trade happens, who stands between the parties, and how quickly it settles all change the price paid and the risk carried.

Learning objectives

By the end of this module, you will be able to:

- 1 **Describe** how exchanges and over-the-counter markets are organised, and apply the four order types.
- 2 **Define** market liquidity, name its five dimensions, and compute spread-based measures of it.
- 3 **Explain** how information becomes price, and distinguish the three forms of market efficiency.
- 4 **Identify** explicit and implicit trading costs, and trace a trade through clearing and settlement.

Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

Topic	What it covers	Part	Source	Serves
1	Stock exchanges: how orders become one price	A • Venues	L11 V1	LO 1
2	The four order types	A • Venues	L11 V1	LO 1
3	Over-the-counter markets	A • Venues	L11 V2	LO 1
4	Market liquidity and its five dimensions	B • Liquidity	L12 V1	LO 2
5	Measuring liquidity	B • Liquidity	L12 V1	LO 2
6	Price discovery and market efficiency	B • Efficiency	L12 V2	LO 3
7	Market structure and trading costs	C • Mechanics	L13 V1	LO 4
8	Margin trading and short selling	C • Mechanics	L13 V1	LO 4
9	Clearing and settlement	C • Mechanics	L13 V2	LO 4

Key terms at a glance



Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

Primary market

Where securities are issued for the first time; proceeds go to the issuer.

Designated market maker

The specialist assigned to a listed stock, quoting both sides and absorbing imbalances.

Market order

An instruction to execute immediately at the best available price.

Stop-loss order

An instruction to sell automatically when the price falls to a trigger level.

Over-the-counter market

A decentralised dealer network where prices are negotiated bilaterally.

Central counterparty

The clearing house that legally inserts itself between every buyer and seller.

Tightness

The cost of immediate execution, captured by the bid-ask spread.

Resiliency

How quickly prices recover after a temporary imbalance.

Flight to liquidity

The rush into safe, tradable assets during a stress event.

Relative spread

Spread divided by midpoint, which standardises across price levels.

Amihud ILLIQ

Average absolute return per rupee of volume; higher means less liquid.

Adverse selection

The dealer's risk of trading against a better-informed counterparty.

Payment for order flow

Rebates brokers receive for routing orders to particular venues.

Percent margin

Equity in the account as a proportion of the value of securities held.

Netting

Aggregating and offsetting buy and sell obligations so only the net position moves.

Secondary market

Where existing securities trade between investors; proceeds go to the seller.

Limit order book

The public record of pending orders at each price level.

Limit order

An instruction to execute only at a specified price or better.

Stop-buy order

An instruction to buy automatically when the price rises through a trigger.

Negotiated Dealing System

The RBI platform on which Indian government securities trade.

Liquidity

The ability to buy or sell quickly, in size, at low cost and with minimal price impact.

Depth

The volume available at the best quoted bid and ask.

Liquidity premium

The extra return demanded for holding assets that are hard to exit.

Quoted spread

Ask minus bid — the round-trip cost of the smallest possible trade.

Effective spread

Twice the absolute gap between fill price and midpoint — the true cost of a fill.

Kyle's lambda

The coefficient linking price change to net order flow; a measure of price impact.

Efficient market hypothesis

Fama's proposition that prices reflect available information, in weak, semi-strong or strong form.

Decimalisation

The 2001 shift to one-cent ticks, which collapsed spreads.

Uptick rule

The restriction permitting short sales only after a positive price change.

T+1 settlement

Settlement one working day after the trade date — India's cycle since January 2023.

PART A • WHERE TRADING HAPPENS

The venues, and how to speak to them

1 Stock Exchanges: How Orders Become One Price

PART A • WHERE TRADING HAPPENS • TOPIC 1 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

An exchange takes every order in the country, routes it to one place, and matches the highest bid against the lowest ask. That match is a price.

Why this matters

Week 1 said secondary markets create liquidity. This week opens the machine and shows how. Everything about liquidity in Part B and settlement in Part C assumes you know what happens at the moment of a trade.

Primary and secondary, one more time

Markets do one profound job: move funds from those with a surplus to those with a shortage. In the **primary market** new securities are issued for the first time and the proceeds go to the issuer. In the **secondary market** existing securities change hands between investors, and the issuer receives nothing.

Primary markets create capital. Secondary markets create liquidity. Secondary trading happens in two venue types: organised exchanges, centralised and regulated, and over-the-counter dealer networks. Both deliver liquidity and price discovery.

Inside an organised exchange

Listing requirements. Not every firm qualifies — listed companies must clear minimum thresholds on revenue, share capital and governance. **Centralised auction.** Every buy and sell order from across the country is routed to one place and matched continuously, thousands of trades per second. **Price formation.** The price settles where the highest bid meets the lowest ask; multiplied at scale, that produces the live market price. **Designated market makers.** Each listed stock has one DMM, also called a specialist, assigned to quote both sides, absorb imbalances and maintain the limit order book. **Oversight.** Every trade is publicly reported; SEBI polices India, the SEC the United States.

India's two exchanges, and the global picture

The **BSE**, established 1875, is Asia's oldest, with over 5,000 listings and the Sensex tracking thirty large-cap stocks. The **NSE**, established 1992, was born electronic — screen-based from day one — carries the Nifty 50, and is the largest equity derivatives exchange in the world, with average daily turnover above ₹40,000 crore.

Abroad, the **NYSE** was founded under the Buttonwood Agreement of 1792 and still operates a physical Wall Street floor with DMMs. **NASDAQ**, established 1971, was the world's first fully electronic exchange, with no floor and several competing market makers per stock rather than a single DMM.

INDIA'S TWO EXCHANGES • HERITAGE AND ELECTRONIC

BSE • BOMBAY STOCK EXCHANGE**ESTABLISHED**

1875 — Asia's oldest

BENCHMARK

Sensex, thirty large-cap stocks

LISTINGS

Over 5,000, including Reliance, TCS, HDFC Bank

PLATFORM

BOLT Plus, fully electronic since 1995

NSE • NATIONAL STOCK EXCHANGE**ESTABLISHED**

1992 — screen-based from day one

BENCHMARK

Nifty 50

LISTINGS

Over 2,000 companies

DISTINCTION

Largest equity derivatives exchange in the world; ₹40,000 crore daily turnover

The NYSE (1792) still runs a physical floor. NASDAQ (1971) never had one. The NSE was electronic from its first day — which is why it never had to make the transition.

Figure 1.1 The two Indian exchanges, and where they sit against the global pair.

KEY IDEA

An exchange is not a building; it is a **matching rule**. Centralise every order, then pair the highest bid with the lowest ask. Everything else — listing rules, market makers, reporting — exists to make that rule trustworthy.

EXAM POINTER

Primary versus secondary should be answered on four axes: what trades, who receives the funds, who the intermediary is, and the regulatory regime. The single most-missed mark is that in a secondary trade the proceeds go to the *seller*, not the issuing company.

QUICK RECAP • TOPIC 1

- Primary markets create capital; secondary markets create liquidity.
- An exchange centralises orders and matches the highest bid to the lowest ask.
- Listing rules, DMMs and public trade reporting make that matching trustworthy.
- BSE (1875) carries the Sensex; NSE (1992) carries the Nifty 50 and leads the world in equity derivatives.
- NYSE dates from 1792 and still has a floor; NASDAQ (1971) was the first fully electronic exchange.

2 The Four Order Types

PART A • WHERE TRADING HAPPENS • TOPIC 2 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Two ways to enter a position and two ways to protect one. Every trading strategy is built from these four.

Why this matters

Order types are the vocabulary of execution. They reappear in Part C when trading costs are measured, and in Week 10 when hedging decisions have to be implemented rather than merely described.

Speed against price

A **market order** executes immediately at the best available price. You accept whatever the market offers right now — speed over price certainty. Use it when getting filled matters more than the exact price paid.

A **limit order** executes only at your specified price or better: buy Infosys at ₹2,500 or lower. Price over speed — and the trade-off is that you may wait, or never fill at all.

The two protective triggers

A **stop-loss order** sells automatically when the price falls to a trigger level. It is the classic tool for capping the loss on a position you already hold.

A **stop-buy order** is the mirror image: it buys automatically when the price rises through a trigger, which is how a trader enters a breakout or momentum position.

Why the distinction matters for cost

A market order guarantees execution but not price, so it crosses the bid-ask spread immediately and pays for immediacy. A limit order guarantees price but not execution, so it supplies liquidity rather than consuming it. That distinction becomes measurable in Topic 5, where the spread turns out to be exactly the cost of demanding immediacy.

FOUR ORDERS • TWO FOR ENTRY, TWO FOR PROTECTION

Order type	What it does	The trade-off	When to use it
Market	Executes immediately at the best available price	Speed over price certainty	When getting filled matters more than the price
Limit	Executes only at your price or better	Price over speed — may never fill	When the price matters more than certainty of execution
Stop-loss	Sells automatically when price falls to a trigger	Locks in a loss to prevent a larger one	Capping downside on a position you hold
Stop-buy	Buys automatically when price rises through a trigger	Enters after the move has begun	Entering a breakout or momentum position

Two execution styles and two protective triggers. Every trading strategy combines these four.

Figure 2.1 The four order types. Note that the first two differ on what they guarantee.

COMMON PITFALL

Assuming a stop-loss guarantees the exit price. It guarantees only that an order is *triggered* at that level; the fill happens at whatever the market then offers. In a fast-falling market the two can differ substantially.

EXAM POINTER

Name all four with what each guarantees and what it sacrifices. The examinable pairing is market versus limit — execution certainty against price certainty — because a candidate who states that clearly has understood why the spread is a cost at all.

QUICK RECAP • TOPIC 2

- Market orders guarantee execution but not price.
- Limit orders guarantee price but not execution.
- Stop-loss orders sell at a trigger to cap downside on a held position.
- Stop-buy orders buy at a trigger to enter on momentum.
- Market orders consume liquidity and pay the spread; limit orders supply it.

3 Over-the-Counter Markets

PART A • WHERE TRADING HAPPENS • TOPIC 3 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

The largest pools of capital in finance — government debt and foreign exchange — never touch an exchange at all.

Why this matters

Students meet exchanges first and assume they are the main event. By volume they are not. Understanding why certain instruments live off-exchange explains a great deal about bond markets in Weeks 7 and 8.

What an OTC market is

Decentralised: no single physical or virtual location; trading happens across dealer networks and electronic platforms worldwide. **Dealer-driven:** securities trade directly between dealers and investors, not on a floor or central limit order book. **Bilateral pricing:** prices are negotiated between two parties rather than set by public auction, with multiple dealers competing for order flow. **Diverse instruments:** government bonds, currencies, unlisted equities and a vast universe of derivatives.

The trade-off is lower transparency in exchange for size, customisation and access to instruments no exchange lists.

India's four OTC segments

Government securities trade entirely OTC between banks and primary dealers, through the RBI's Negotiated Dealing System, with daily turnover crossing ₹30,000 crore and settlement through Subsidiary General Ledger accounts at the RBI. **Foreign exchange** is the world's largest OTC market at \$7.5 trillion daily globally, with no central exchange anywhere; dealer banks including SBI, HDFC, Citigroup and Deutsche Bank quote two-way prices around the clock. **Over-the-counter equity** serves firms outside main listings — NSE Emerge and BSE SME in India, and the OTCQX, OTCQB and Pink Sheets tiers in the United States. **Interest rate swaps** let banks and corporates hedge rate risk bilaterally, exchanging fixed for floating cash flows on a notional principal.

NASDAQ, the hybrid, and the 2008 reform

NASDAQ is famous as an exchange but is technically an OTC quotation network: dealers hold inventory, many market makers compete on each stock, and discovery comes from competition rather than auction. Three reforms reshaped it — the Small Order Execution System of 1987, electronic communication networks such as Instinet and Archipelago, and the 1997 display rules requiring all customer limit orders to be shown. The result was tighter spreads and cleaner price formation.

After 2008, regulators worldwide mandated **central clearing** of standardised OTC derivatives. Bilateral contracts had created webs of opaque counterparty exposure — when Lehman fell, nobody knew who else was on the hook. A central counterparty now steps between every trade, netting and guaranteeing performance. Bespoke contracts remain bilateral, though margin requirements have tightened.

EXCHANGE AND OVER-THE-COUNTER • THE SHARPEST CONTRAST IS TRANSPARENCY

STOCK EXCHANGE
LOCATION

Centralised

PRICE DISCOVERY

Public auction, bid and ask on the book

TRANSPARENCY

High — all trades publicly reported

REGULATION

Strict listing and trading rules

PARTICIPANTS

Retail and institutional

OVER-THE-COUNTER MARKET
LOCATION

Decentralised dealer networks

PRICE DISCOVERY

Bilateral negotiation

TRANSPARENCY

Lower — limited pre-trade visibility

REGULATION

Lighter, with few listing rules

PARTICIPANTS

Primarily institutional

Exchanges broadcast everything. OTC markets keep prices bilateral and oversight lighter — which is exactly why they can carry instruments an exchange could never standardise.

Figure 3.1 The seven-dimension comparison, condensed to the five that carry marks.

KEY IDEA

Volume and visibility are not the same thing. Foreign exchange turns over \$7.5 trillion a day with no exchange anywhere — more than the combined annual output of several large economies, traded entirely through bilateral dealer relationships.

EXAM POINTER

Comparison questions want at least five axes: location, price discovery, transparency, regulation and participants. Name all four Indian OTC segments if asked about the domestic market, and state that government securities trade through the RBI's Negotiated Dealing System — that specific detail separates a good answer from a generic one.

QUICK RECAP • TOPIC 3

- OTC markets are decentralised dealer networks with bilaterally negotiated prices.
- They trade lower transparency for size, customisation and instruments no exchange lists.
- India's four segments: government securities, foreign exchange, unlisted equity, interest rate swaps.
- NASDAQ is technically an OTC quotation network, reshaped by SOES, ECNs and the 1997 display rules.
- Since 2008, standardised OTC derivatives must clear centrally through a central counterparty.

PART B • LIQUIDITY AND EFFICIENCY

The two properties that make a market work

4 Market Liquidity and Its Five Dimensions

PART B • LIQUIDITY AND EFFICIENCY • TOPIC 4 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

Liquidity is the ability to buy or sell quickly, in size, at low cost and with minimal price impact — and it is invisible until it disappears.

Why this matters

In September 2008 the bid-ask spread on corporate bonds went from a few hundredths of a percent to several full percentage points. Institutions holding billions found they could not sell at any price. That was not a price problem. It was a liquidity problem.

What liquidity actually is

Sell a thousand shares of Reliance at half past ten in the morning and it takes seconds, near the last quoted price. Sell a thousand shares of an obscure stock and it may mean hours of waiting, a sharp price drop, and costs that swallow the return.

The deeper point is that realised returns depend not only on expected returns but on **execution costs**. Liquidity converts a paper return into a realised one — what you keep depends on how cleanly you can exit — which makes it a core determinant of investment attractiveness rather than a side issue.

Five dimensions

Tightness: the cost of immediate execution, captured by the bid-ask spread. **Depth**: the volume available at the best quoted bid and ask. **Breadth**: order volumes spread across multiple price levels. **Immediacy**: the speed at which a trade can be filled. **Resiliency**: how quickly prices recover after a temporary imbalance.

Only when all five are strong can a market be called truly liquid. A market can be tight at the top and still shallow one level down.

Liquidity risk and asset pricing

Illiquid assets must reward investors for the difficulty of getting in and out cleanly. Higher illiquidity means higher required return — the **liquidity premium** — and it shows up in small-cap stocks, corporate bonds, emerging market securities and private credit. Patient capital able to hold through illiquid windows captures it.

The reverse appears under stress. A **flight to liquidity** follows a bank failure, sovereign shock or volatility spike: spreads widen sharply, depth evaporates, dealers pull back quotes, and capital rushes into treasury securities. In stressed markets the ability to exit can matter more than the position itself.

FIVE DIMENSIONS • A MARKET IS LIQUID ONLY WHEN ALL FIVE HOLD

Tightness

The cost of immediate execution, measured by the bid-ask spread.

Depth

Volume available at the best quoted bid and ask.

Breadth

Order volumes spread across multiple price levels.

Immediacy

How fast a trade can actually be filled.

Resiliency

How quickly prices recover after a temporary imbalance.

All five together

Strength in one dimension does not compensate for weakness in another.

*Liquidity is **invisible when it is there**. You notice it only when it is gone — and when it goes, as 2008 showed, it can bring down entire financial systems in days.*

Figure 4.1 The five dimensions of liquidity. Each measures something the others do not.

REAL-WORLD EXAMPLE

After Lehman failed in September 2008 the market was technically open — you could place a trade. But corporate bond spreads exploded and there were simply no buyers. The Federal Reserve responded not by cutting rates but by injecting \$1.5 trillion specifically to restore the ability to trade.

EXAM POINTER

Name all five dimensions — partial lists lose marks proportionally — and give one line on what each measures. Then state the liquidity premium relationship: higher illiquidity, higher required return. A question on crisis behaviour wants flight to liquidity named explicitly.

QUICK RECAP • TOPIC 4

- Liquidity is the ability to buy or sell quickly, in size, at low cost, with minimal price impact.
- Realised returns depend on execution costs, so liquidity is a core determinant of attractiveness.
- Five dimensions: tightness, depth, breadth, immediacy, resiliency.
- Illiquid assets pay a liquidity premium; patient capital captures it.
- Under stress, capital flees to liquid assets and the ability to exit outweighs the position itself.

5 Measuring Liquidity

PART B • LIQUIDITY AND EFFICIENCY • TOPIC 5 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

The spread is the most direct measure, but no single number captures liquidity — which is why professionals triangulate across several.

Why this matters

This is the quantitative core of the week. The spread measures reappear as trading costs in Part C, and Kyle's lambda returns in Week 13 when derivatives desks model price impact.

The bid-ask spread

The **bid** is the highest price a buyer will pay; the **ask** is the lowest a seller will accept. The gap between them is the cost of immediacy, and it is the dealer's compensation.

A narrow spread means low cost, high competition among participants and high liquidity. A wide spread means high cost, few competing dealers and thin liquidity.

Why spreads exist at all

Three cost sources. **Order processing**: the administrative, infrastructure and operational cost of matching, clearing and settling each trade, which dealers must recover. **Inventory holding**: dealers hold positions to provide immediacy and bear price risk while doing so. **Adverse selection**: the risk of trading against an informed counterparty who knows something the dealer does not — to avoid being picked off, dealers widen quotes.

The spread is not a coincidence. It pays for processing, risk bearing and information asymmetry.

Reading the limit order book

A snapshot might show 10,000 shares bid at ₹1,000 and 8,000 bid at ₹1,001, against 9,000 offered at ₹1,002 and 12,000 at ₹1,003. The best bid is ₹1,001 and the best ask ₹1,002, so the spread is ₹1 — a tight top. Resting orders stacked at successive levels cushion price gaps, so large trades fill with little slippage. Thin levels do the opposite: modest orders jump the price.

THREE SPREAD MEASURES ON ONE QUOTE

Measure	Formula	Worked on bid ₹998 / ask ₹1,002	What it is for
Quoted spread	Ask – bid	$1,002 - 998 = ₹4$	The round-trip cost of the smallest possible trade
Midpoint	$(\text{Ask} + \text{bid}) \div 2$	$(1,002 + 998) \div 2 = ₹1,000$	The reference price for the other two
Relative spread	$\text{Spread} \div \text{midpoint}$	$4 \div 1,000 = 0.4\%$	Standardises across price levels — compares a ₹50 stock with a ₹50,000 one
Effective spread	$2 \times \text{trade price} - \text{midpoint} $	Depends on the actual fill	The true cost when a fill drifts off midpoint; can exceed the quoted spread

Anyone who buys and immediately sells loses **₹4 per share**. That is not a fee — it is the price of demanding immediacy, and it is the dealer's compensation.

Figure 5.1 The three spread measures. Relative spread is the one that allows comparison across securities.

KEY IDEA

No single measure tells the whole story. Combine a spread measure, a volume measure and an impact measure and you triangulate real liquidity; rely on one and you will misjudge it.

REAL-WORLD EXAMPLE

Three further measures widen the picture. **Turnover ratio** is volume divided by shares outstanding — higher means deeper participation. **Amihud ILLIQ** averages absolute return per rupee of volume — higher means prices move more per unit traded, so less liquidity. **Kyle's lambda** is the price impact coefficient on order flow — a large lambda means even modest flow moves prices strongly.

EXAM POINTER

Compute, do not describe. Given a bid and ask, produce the quoted spread, the midpoint and the relative spread with the arithmetic shown. Then state the three reasons spreads exist — order processing, inventory, adverse selection — because the explanation marks sit there, not in the calculation.

QUICK RECAP • TOPIC 5

- The bid-ask spread is the most direct measure and represents the cost of immediacy.
- Quoted spread = ask – bid; relative spread = spread ÷ midpoint, which standardises across price levels.
- Effective spread captures the true cost of an actual fill and can exceed the quoted spread.
- Spreads exist for three reasons: order processing, inventory holding, adverse selection.
- Turnover, Amihud ILLIQ and Kyle's lambda measure different facets; triangulate rather than rely on one.

6 Price Discovery and Market Efficiency

PART B • LIQUIDITY AND EFFICIENCY • TOPIC 6 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

Because dealers cannot tell an informed trader from a noise trader, they widen spreads — which makes the spread itself a measure of hidden information.

Why this matters

Week 1 introduced price discovery descriptively. This topic supplies the mechanism, and the efficient market hypothesis that follows from it governs every claim about beating the market in Weeks 5, 6 and 11.

How information becomes price

Four sources drive prices towards fundamentals. **Public announcements** — corporate disclosures, regulatory filings, press releases — reset what every participant knows simultaneously. **Earnings reports** trigger moves within seconds. **Macroeconomic data** such as RBI rate decisions, inflation prints and GDP releases shift every asset at once. And **trading itself** reveals information: a large buy order may signal private knowledge the market does not yet have.

Information asymmetry and its four consequences

Adverse selection: dealers fear whoever trades against them may know more, so every quote risks being picked off. **Spreads widen** to protect against informed flow — the wider the spread, the higher the hidden information risk. **Depth falls:** fewer shares are quoted at the best prices and books thin out. **Retail disadvantage:** investors on Zerodha or Groww trade against participants who may know far more, and pay for it in worse fills.

Two models worth knowing

The Kyle model (1985) formalises how order flow carries information. It has three participants: an *informed trader* holding private information and sizing orders to disguise intent; *noise traders* whose random flow provides the cover the informed trader hides behind; and a *market maker* who observes net flow but not its source. The result is $\Delta P = \lambda \times Q$ — price change equals the price impact coefficient times signed order flow.

Glosten-Milgrom explains the spread itself. The dealer cannot distinguish informed from uninformed traders; anyone at the quote could be an insider. Every trade therefore carries the possibility of private information, so the dealer widens the spread to avoid being picked off. Greater asymmetry means wider spreads — which makes the spread a signal of how much hidden information is circulating, not merely a transaction cost.

THE EFFICIENT MARKET HYPOTHESIS • EACH FORM RULES OUT A DEEPER CLASS OF EDGE

Form	What prices reflect	What cannot win	Status
Weak	All historical prices	Technical analysis — chart patterns hold no predictive edge	The weakest claim, most easily defended
Semi-strong	All public information: earnings, balance sheets, news	Fundamental analysis — annual reports are priced in before you read them	The most empirically tested form
Strong	All information, public and private	Even corporate insiders earn no abnormal return	Rarely holds — which is why insider trading laws exist

*Liquidity **supports** efficiency but cannot guarantee it. Noise trading, fragmented venues, algorithmic distortions and hidden orders in dark pools all distort even deep markets.*

Figure 6.1 Fama's three forms. Each stronger version rules out a deeper class of trading edge.

KEY IDEA

The bid-ask spread is doing two jobs at once. It is a cost you pay, and it is a **measurement** of how much private information the dealer believes is circulating. Wide spreads are informative, not merely expensive.

EXAM POINTER

Name all three EMH forms with what each rules out. For the microstructure models, state the Kyle formula $\Delta P = \lambda \times Q$ and name all three participants, then give Glosten-Milgrom's conclusion in one sentence. A question on why liquidity does not guarantee efficiency wants at least two of the four distortions named.

QUICK RECAP • TOPIC 6

- Information reaches prices through announcements, earnings, macro data and trading itself.
- Asymmetry produces four consequences: adverse selection, wider spreads, falling depth, retail disadvantage.
- Kyle (1985): $\Delta P = \lambda \times Q$ — order flow itself carries a signal.
- Glosten-Milgrom: dealers widen spreads because they cannot identify informed traders.
- EMH comes in weak, semi-strong and strong forms; liquidity supports efficiency but cannot guarantee it.

PART C • TRADING AND SETTLEMENT

From order to irrevocable ownership

7 Market Structure and Trading Costs

PART C • TRADING AND SETTLEMENT • TOPIC 7 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

The same share can trade on an exchange, through dealers, or bilaterally — and the structure determines the price you actually pay.

Why this matters

Part C moves from what a market is to what it costs to use one. The implicit costs here are often larger than the commission, and they are the ones learners consistently overlook.

Three structures

Market structure is the set of rules, participants and mechanisms through which securities are bought and sold in a venue. It determines how prices form, how efficiently trades execute, and the total cost.

National and regional exchanges are organised, centralised venues where only members may trade and each listed stock has a designated market maker. **OTC dealer networks** are decentralised, with competing dealers quoting bids and asks and earning the spread. **Direct bilateral trading** happens institution to institution, without intermediaries, for sizeable blocks traded privately off exchange.

Structure equals outcome: the same Infosys share traded through each route produces a slightly different price.

Explicit and implicit costs

Commission is the visible cost — what the broker charges, fully disclosed. Full-service brokers provide advice, research and relationship management at higher fees; discount brokers such as Zerodha and Groww offer low-cost execution with no advisory layer.

The implicit costs are larger and less obvious. The **bid-ask spread** is crossed on both entry and exit. **Price impact** means a large order moves the market against the trader before the fill completes. **Payment for order flow** means brokers receive rebates for routing orders to particular venues, which raises a genuine conflict of interest.

What reduced costs

Decimalisation in 2001 replaced one-eighth-dollar minimum ticks with one-cent ticks; spreads collapsed and trading costs fell sharply, at the expense of dealer revenues. The **Intermarket Trading System** links exchanges by computer and routes orders to the best available venue, so brokers must use the best price. **Electronic communication networks** display best limit orders and match algorithmically — the trade-off being market fragmentation and weaker after-hours price discovery.

WHERE A TRADE CAN HAPPEN • AND WHAT EACH VENUE IS FOR

	Exchanges	NASDAQ	ECNs
Price formation	Centralised specialist auction	Dealer competition	Algorithmic matching
Transparency	High — visible book	Medium	Low — hidden orders
Trading hours	Fixed session	Extended via SOES	Around the clock
Market makers	One DMM per stock	Multiple competing	Algorithmic or none
Typical spread	Low	Historically high, narrow after 1997	Often near zero
Best suited for	Listed equity discovery	Retail and dealer flow	Institutional crossing

Competition between the NYSE, NASDAQ and ECNs for order flow has driven **spreads down and fill speeds up**. The investor gained; the dealer's margin did not.

Figure 7.1 Three venue types compared. Note that transparency and spread run in opposite directions.

COMMON PITFALL

Comparing brokers on commission alone. A zero-commission broker earning payment for order flow may route your order to a venue giving a worse fill — and the implicit cost of that worse fill can exceed the commission you saved.

EXAM POINTER

Separate explicit from implicit costs and name at least two of each. The three implicit costs — spread, price impact, payment for order flow — are where the marks sit, because they are the ones a naive answer misses. Add decimalisation with its date, 2001.

QUICK RECAP • TOPIC 7

- Market structure determines how prices form, how efficiently trades execute and what they cost.
- Three structures: organised exchanges, OTC dealer networks, direct bilateral trading.
- Explicit cost is commission; implicit costs are the spread, price impact and payment for order flow.
- Implicit costs are often larger than the commission.
- Decimalisation (2001), the Intermarket Trading System and ECNs all narrowed spreads.

8 Margin Trading and Short Selling

PART C • TRADING AND SETTLEMENT • TOPIC 8 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

Both let a trader take a position larger or more contrarian than their cash allows — and both can lose more than the amount committed.

Why this matters

These two mechanisms reappear throughout the derivatives weeks. Margin is the foundation of futures in Week 10; short selling is what makes bearish views expressible at all.

Buying on margin

The investor borrows part of the purchase price from the broker, funded through broker's call loans. **Initial margin** is set at 50% by the US Federal Reserve Board. **Percent margin** is equity in the account divided by the value of securities held. When it falls below the maintenance threshold, the broker issues a **margin call**.

Short selling

The investor **borrow**s shares from the broker's inventory or another lender, **sells** them immediately at the prevailing price expecting a fall, **pays dividends** to the lender during the borrow period while posting margin throughout, and finally **covers** by buying the shares back and returning them.

The **uptick rule** permits short sales only after a positive price change, to prevent coordinated selling from accelerating a falling stock.

The asymmetry that matters

A buyer's loss is capped at the amount invested — a share can only fall to zero. A short seller's loss is theoretically **unlimited**, because the price can rise without bound. Borrowing limits, margin rules and exchange controls all constrain the trade.

Set against that, short selling lets bearish views enter the market and improves price discovery. A market in which only optimism can be expressed prices badly.

LEVERAGE CUTS BOTH WAYS • A 30% FALL HALVES THE EQUITY

Stage	Position value	Loan	Equity	Percent margin
Purchase	₹1,00,000	₹40,000	₹60,000	60%
Price falls 30%	₹70,000	₹40,000 (unchanged)	₹30,000	About 43%
Consequence	—	—	Halved	May breach the maintenance threshold

The loan does not shrink when the position does. A 30% fall in the share price cuts equity in half, and a margin call forces a sale at the worst possible moment.

Figure 8.1 Margin arithmetic. Note that the borrowed amount is fixed while the equity absorbs the entire loss.

KEY IDEA

Leverage does not change the size of a price move. It changes what proportion of *your* money that move represents — which is why a modest fall in the security can be a severe loss in the account.

EXAM POINTER

Percent margin = equity ÷ value of securities. Compute it at both stages and show that the loan is constant. For short selling, name all four steps in order and state the asymmetry explicitly: capped loss for the buyer, theoretically unlimited for the short.

QUICK RECAP • TOPIC 8

- Margin trading borrows part of the purchase price; initial margin is 50% under US rules.
- Percent margin = equity ÷ value of securities; breaching the maintenance level triggers a margin call.
- A 30% price fall on a ₹1,00,000 position with ₹40,000 borrowed halves the equity.
- Short selling borrows, sells, pays dividends to the lender, then covers.
- A buyer's loss is capped; a short seller's is theoretically unlimited.

9

Clearing and Settlement

PART C • TRADING AND SETTLEMENT • TOPIC 9 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

Five steps stand between a trade being agreed and ownership legally changing hands — and a central counterparty guarantees every one of them.

Why this matters

This is the infrastructure that makes it possible to trade with a stranger anywhere in the world without thinking about whether they will deliver. In November 1987 it very nearly failed.

The five steps

Trade execution. Buyer and seller agree price and quantity, often through brokers or DMMs, and the trade is recorded on a timestamped file the instant it executes — the Consolidated Tape in the United States.

Trade matching. Both brokers independently confirm details to the clearing house. Security identifier, trade price and counterparty identity must match exactly; any mismatch generates a break report and the trade is held until resolved.

Clearing and netting. All buy and sell obligations for each stock are aggregated and offset, drastically shrinking the volume of cash and shares that must actually move. The central counterparty legally inserts itself between every buyer and seller.

Settlement. The purchaser delivers cash and the seller delivers securities. India has settled at T+1 since January 2023, with NSDL and CDSL holding shares electronically in demat accounts.

Finality. The transfer becomes legally irrevocable, and if one party defaults the clearing house honours the trade itself.

Why netting matters

Suppose a broker's clients buy 1,00,000 Infosys shares and sell 95,000 on the same day. Without netting, 1.95 lakh shares would have to move. With netting, just 5,000 change hands — the net position. Multiply that across every security and every broker and the reduction in settlement traffic is enormous.

The regulatory bedrock, and the guardrails

The **Securities Act of 1933** governs the primary market: full disclosure of material information when securities are offered publicly, through a prospectus filed with the regulator. Its principle is to give investors the facts and let them decide. The **Securities Exchange Act of 1934** governs the secondary market, created the SEC itself, requires every broker and dealer to register, and underpins enforcement against fraud and manipulation.

Two guardrails sit on top. **Insider trading** — trading on material, non-public information — is criminal, with penalties including fines, disgorgement and imprisonment; the definition of an insider is deliberately broad, since suppliers, analysts and consultants may all receive material information. **Circuit breakers** halt trading when an index falls by preset amounts, letting the market absorb new information calmly rather than cascading.

HOW SETTLEMENT GOT FASTER • EACH COMPRESSION REMOVES A DAY OF RISK

Period	Cycle	Market
Pre-1995	T+5	United States — five working days between trade and settlement
1995–2017	T+3	United States — shortened, but still days of counterparty exposure
2017–2024	T+2	United States — two-day cycle across major markets
January 2023	T+1	India — the NSE moved to one-day settlement, among the fastest globally
2024 onwards	T+1	United States — joins India at one-day finality

Each compression of the cycle **removes a day of counterparty risk** from the entire financial system. India reached T+1 a year before the United States.

Figure 9.1 The settlement cycle, shortening. The direction of travel is one way.

KEY IDEA

The central counterparty is what makes anonymous trading possible. Because it becomes the buyer to every seller and the seller to every buyer, no participant needs to assess any other participant's creditworthiness — counterparty risk becomes a system-wide non-issue.

REAL-WORLD EXAMPLE

After Black Monday in October 1987, brokers who had sold shares could not deliver them and buyers who had paid could not receive what they had bought. The clearing and settlement infrastructure came close to seizing up entirely, and the Federal Reserve intervened not only to stabilise prices but to guarantee that trades would settle.

EXAM POINTER

Name all five steps in order — execution, matching, netting, settlement, finality — and explain netting with a numerical example. State India's T+1 date, January 2023. For the Acts, pair each with its market: 1933 primary, 1934 secondary and the creation of the SEC.

QUICK RECAP • TOPIC 9

- Five steps: execution, matching, clearing and netting, settlement, finality.
- Netting offsets buys against sells so only the net position moves.
- The central counterparty inserts itself between every buyer and seller, eliminating default risk.
- India has settled at T+1 since January 2023; NSDL and CDSL hold shares electronically.
- The 1933 Act governs new issues; the 1934 Act governs secondary trading and created the SEC.

Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

LO	What you should now be able to do	Where it was covered
1	Describe exchange and OTC organisation and apply the four order types	Topics 1, 2 and 3 • Figures 1.1, 2.1, 3.1
2	Define liquidity, name its dimensions and compute spread measures	Topics 4 and 5 • Figures 4.1, 5.1
3	Explain how information becomes price and distinguish the three forms of efficiency	Topic 6 • Figure 6.1
4	Identify trading costs and trace a trade through settlement	Topics 7, 8 and 9 • Figures 7.1, 8.1, 9.1

The module in six sentences

An exchange is a matching rule: centralise every order and pair the highest bid with the lowest ask. Investors speak to it through four order types, two of which guarantee execution and two of which guarantee price. The largest markets in finance — government debt and foreign exchange — never touch an exchange at all, trading bilaterally through dealer networks with lower transparency but far greater size and flexibility. Liquidity has five dimensions and is invisible until it vanishes, which is why the spread, the turnover ratio and the price impact coefficient must be read together rather than singly. Prices absorb information through announcements, earnings, macro data and trading itself, and because dealers cannot tell informed traders from noise traders they widen spreads — making the spread a measure of hidden information as well as a cost. A trade then passes through five steps to finality, with netting shrinking what actually moves and a central counterparty guaranteeing that it moves at all.

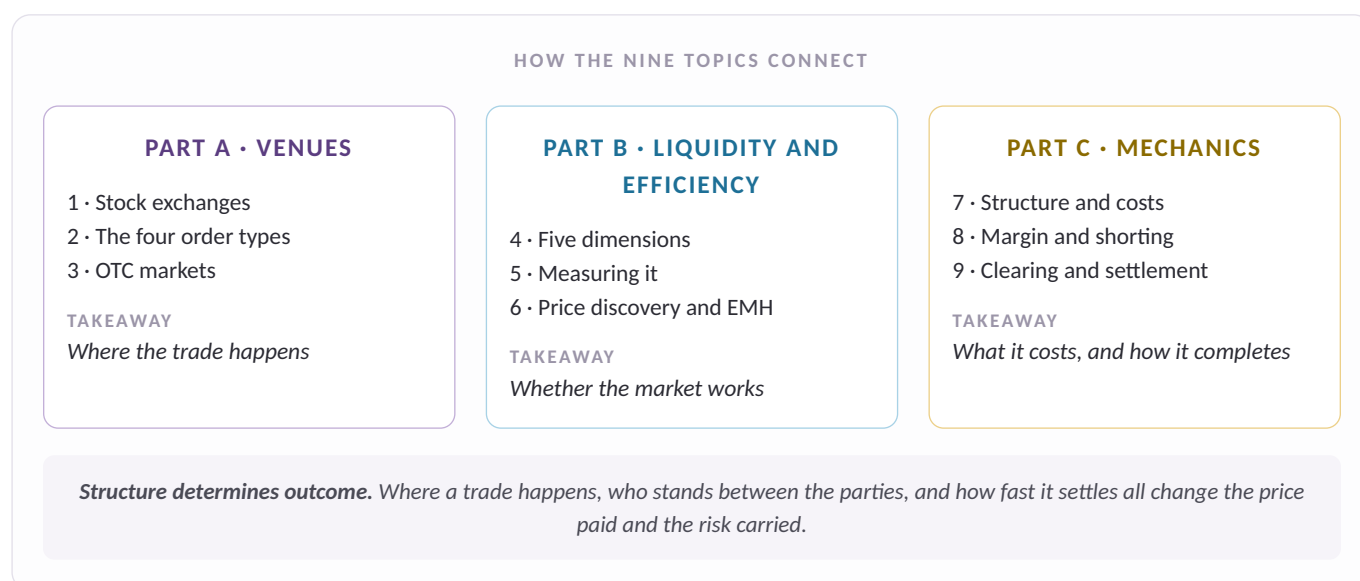


Figure M1 How the nine topics connect. Revise along the parts, not down the list.

Glossary and self-check



Term	Definition
Primary market	Where securities are issued for the first time; proceeds go to the issuer.
Secondary market	Where existing securities trade between investors; proceeds go to the seller.
Designated market maker	The specialist assigned to a listed stock, quoting both sides and absorbing imbalances.
Limit order book	The public record of pending orders at each price level.
Market order	An instruction to execute immediately at the best available price.
Limit order	An instruction to execute only at a specified price or better.
Stop-loss order	An instruction to sell automatically when the price falls to a trigger level.
Stop-buy order	An instruction to buy automatically when the price rises through a trigger.
Over-the-counter market	A decentralised dealer network where prices are negotiated bilaterally.
Negotiated Dealing System	The RBI platform on which Indian government securities trade.
Central counterparty	The clearing house that legally inserts itself between every buyer and seller.
Liquidity	The ability to buy or sell quickly, in size, at low cost and with minimal price impact.
Tightness	The cost of immediate execution, captured by the bid-ask spread.
Depth	The volume available at the best quoted bid and ask.
Resiliency	How quickly prices recover after a temporary imbalance.
Liquidity premium	The extra return demanded for holding assets that are hard to exit.
Flight to liquidity	The rush into safe, tradable assets during a stress event.
Quoted spread	Ask minus bid — the round-trip cost of the smallest possible trade.
Relative spread	Spread divided by midpoint, which standardises across price levels.
Effective spread	Twice the absolute gap between fill price and midpoint — the true cost of a fill.
Amihud ILLIQ	Average absolute return per rupee of volume; higher means less liquid.
Kyle's lambda	The coefficient linking price change to net order flow; a measure of price impact.
Adverse selection	The dealer's risk of trading against a better-informed counterparty.
Efficient market hypothesis	

Term	Definition
	Fama's proposition that prices reflect available information, in weak, semi-strong or strong form.
Payment for order flow	Rebates brokers receive for routing orders to particular venues.
Decimalisation	The 2001 shift to one-cent ticks, which collapsed spreads.
Percent margin	Equity in the account as a proportion of the value of securities held.
Uptick rule	The restriction permitting short sales only after a positive price change.
Netting	Aggregating and offsetting buy and sell obligations so only the net position moves.
T+1 settlement	Settlement one working day after the trade date — India's cycle since January 2023.

Self-check

Ungraded retrieval practice. Answer from memory first, then check.

Q1 A stock shows a bid of ₹998 and an ask of ₹1,002. Give the quoted spread, the midpoint and the relative spread.

Spread ₹4; midpoint ₹1,000; relative spread 0.4%. Spread = 1,002 – 998. Midpoint = (1,002 + 998) ÷ 2. Relative spread = 4 ÷ 1,000. The relative figure is what lets you compare this with a ₹50 stock.

Q2 You buy ₹1,00,000 of shares with ₹40,000 borrowed. The position falls 30%. What happens to your percent margin?

It falls from 60% to about 43%. Equity starts at 1,00,000 – 40,000 = ₹60,000, so margin is 60%. After the fall the position is ₹70,000 but the loan is unchanged, so equity is ₹30,000 and margin is 30,000 ÷ 70,000. The loan does not shrink with the position — equity absorbs the whole loss.

Q3 Why is a wide bid-ask spread informative as well as expensive?

Because dealers widen spreads when they fear informed trading. Under Glosten-Milgrom, the dealer cannot tell an insider from a noise trader, so every quote risks being picked off. A wide spread therefore signals that the dealer believes a lot of private information is circulating.

Q4 Clients buy 1,00,000 shares and sell 95,000 of the same stock on one day. How many shares actually settle, and why does it matter?

5,000. Netting offsets buys against sells so only the net position moves, instead of 1.95 lakh shares. Across every security and broker this reduces settlement traffic enormously and cuts the operational risk in the system.

Q5 Technical analysis stops working. Which form of market efficiency does that support, and what would still be possible?

Weak-form efficiency. If historical prices are fully reflected, chart patterns hold no predictive edge. Fundamental analysis might still work, since weak form makes no claim about public information — that would require the semi-strong form.

Q6 Name one explicit and two implicit trading costs, and say which is usually larger.

Explicit: commission. Implicit: the bid-ask spread and price impact (payment for order flow is a third). The implicit costs are usually larger, which is why comparing brokers on commission alone is misleading.